MIL-E-60591A(MU)
24 September 1971
SUPERSEDING
MIL-E-60591(MU)
18 September 1966

MILITARY SPECIFICATION
EJECTOR, PARACHUTE, CARTRIDGE ACTUATED, XM233 ASSEMBLY

1. SCOPE

1.1 This specification covers Ejector Parachute, Cartridge Actuated, XM233, Assembly, loaded with two M91 Cartridge Assemblies, for use in conjunction with ballistically deployed troop chest reserve-type parachute system.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	-	Ammunition and Special Weapons, General Specification For

STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions

DRAWINGS

Munitions Command

F10536009	-	Ejector, Parachute, Cartridge Actuated, XM233 Assembly
C9215469	-	Carton, Package, Ammunition, for Parachute Ejector, XM233
C9215470	-	Box, Package, Ammunition for Parachute Ejector, XM233
IEL-10536009	-	Inspection Equipment List for Ejector, Parachute Cartridge Actuated, XM233 Assembly

FSC 1377

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(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENT

3.1 General - Material, parts, and assemblies shall comply with the requirements specified in the applicable drawings and specifications.

3.2 Firing pin release - The firing pin of the assembled initiator less cotter pin, cartridge and chamber shall release from the cocked position when the application of a load of 15 to 30 pounds has withdrawn the initiator pin .370 inches, minimum and .425 inches, maximum.

3.3 Torque requirement

3.3.1 The chamber and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation.

3.3.2 The seating or bottoming torque for the initiator into the orifice shall be 200 ± 20 inch pounds.

3.3.3 The seating or bottoming torque for the orifice into the manifold shall be 200 ± 20 inch pounds.

3.3.4 The seating or bottoming torque for the outer tube into the manifold shall be 200 ± 20 inch pounds.

3.3.5 The seating or bottoming torque for the seating pin into the inner tube shall be 200 ± 20 inch pounds.

3.4 Functional

3.4.1 Ejectors loaded with accepted cartridges and temperature condition shall meet the following requirements:

3.4.2 Both initiators loaded with accepted cartridges shall not fail to fire.

3.4.3 The ejector assembly shall function without mechanical failure (see 6.2)

3.4.4 The ejector assembly fired under load (see 6.3) shall perform as follows:

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3.4.4.1 The ejector assembly shall be capable of propelling a 10.75 lbs. weight, which is initially restrained by a 70 lbs. force through the ejector stroke.

3.4.4.2 The velocity of the mass upon separation from the inner tube shall be from 45 to 85 feet per second.

3.4.4.3 The thrust for the first peak shall be from 600 to 1000 lbs. and for the second peak from 500 to 1200 pounds.

3.4.5 The ejector shall comply with all the ballistic requirements throughout the temperature range of minus 65 degrees to plus 200 degrees Fahrenheit (F).

3.4.6 The second initiator shall be fired after 2 inches ± ½ inches of inner tube and projected weight travel.

3.5 Workmanship - The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Burr - No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.2 Foreign matter - No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions - Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection - Inspection shall be in accordance with MIL-A-2550, except as specified herein.

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4.2 Inspection provisions

4.2.1 Lot

4.2.1.1 Submission of product - The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot - A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with accepted cartridges.

4.2.2 Examination - One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table I. The total of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table I. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burrs and foreign matter.

TABLE I

Lot Size	Sample Size	Major Accept	Major Reject	Minor Accept	Minor Reject
1 - 300	100% Examination	-	-	-	-
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3
1801 - 10,000	200	2	3	3	4

4.2.2.1 Cartridge, Impulse M91, Drawing D8597230, covering a detail of Drawing F10536009.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Foreign matter	Visual

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4.2.2.1 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Case scratched, wrinkled, bulged or dented	Visual
202. Varnish missing or inadequate	Visual

4.2.2.2 Assembly, Drawing F10536009, less all drawings except D10520407, C10520414 and D859230.

CRITICAL	
1. Cartridge not properly seated	Visual
2. Filter missing or insecure	Visual
3. Nylon patch missing or improperly applied	Visual
4. Improper assembly	Visual
5. Incomplete assembly	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.2.2.3 Assembly, Drawing F10536009, less all drawings except C10520410, C10520411, and MS-19060-3.

CRITICAL	
1. Firing Pin protrusion	Gage
2. Steel ball missing	Gage
3. Striker hole in firing pin housing obstructed	Visual
4. Improper assembly	Visual
5. Incomplete assembly	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined.	

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4.2.2.4 Assembly, Drawing F10536009, less all drawings except C11730531, B10520412, MS19060-3, C10520415, 10520410, 10535978, and MS 24665-155.

CRITICAL

1. Firing Pin not cocked	Visual/Manual
2. Any part missing	Visual
3. Improper assembly	Visual

MAJOR

101. Cotter pin loose or deformed	Manual/Visual
102. Burr	Visual
103. Foreign matter	Visual

MINOR

None defined.

4.2.2.5 Assembly, Drawing F10536009,less all drawings except 10535978, 10520412, 10520409, MS-24665-155, 10535979, 10520415, 10520496, 10520407, 10520414, 10520410, 10520411, and MS-19060-3.

CRITICAL

1. Cartridge face not seated against base of firing pin housing.	X-Ray
2. Any part missing	X-Ray
3. Improper assembly	X-Ray
4. Marking missing, incorrect or illegible	Visual

MAJOR

101. Set screw missing or insecure	Visual
102. Burr	Visual
103. Foreign matter	Visual

MINOR

201. Thread sealing compound missing from set screws	Visual
202. Protective finish inadequate or defective	Visual

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4.2.2.6 Assembly Drawing F10536009

CRITICAL

1.	Any part missing	Visual
2.	Improper assembly	Visual
3.	Marking missing, incorrect or illegible	Visual

MAJOR

101.	Burr	Visual
102.	Foreign matter	Visual

MINOR

201.	Length from initiator to Buffer	Gage
202.	Total length	Gage

4.2.2.7 Packing in Carton Ammunition, Drawing C9215469

CRITICAL

None defined.

MAJOR

None defined.

Defect		Method of Inspection

MINOR

201.	Contents loose	Manual
202.	Cut or damaged	Visual
203.	Markings missing, incorrect or illegible	Visual

4.2.2.8 Packing in Wooden Box, Drawing C9215470

CRITICAL

None defined.

MAJOR

None defined.

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4.2.2.8 (Cont'd)

	Defect	Method of Inspection
MINOR		
201.	Contents loose	Manual
202.	Hardware missing, loose, broken or improperly engaged	Visual and Manual
203.	Strapping missing, loose, broken or improperly assembled	Visual and Manual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking missing, incorrect or illegible	Visual
207.	Number of units incorrect	Visual

4.2.3 Inspection Testing - The following tests shall be performed in accordance with 4.3.

4.2.3.1 Firing pin release - The initiator pin of each subassembly consisting of all initiator parts except the cotter pin, cartridge and chamber, shall be subjected to the firing pin release test. Failure of any subassembly to comply with the requirements of 3.2 shall be cause for rejection of the subassembly.

4.2.3.2 Torque inspection -

4.2.3.2.1 Each initiator subassembly shall be subjected to the breakaway torque application. Failure of any initiator assembly to comply with the requirement of 3.3.1 shall be cause for rejection of the subassembly.

4.2.3.2.2 Periodic unscheduled monitoring shall be conducted for requirements of paragraphs 3.3.2, 3.3.3, 3.3.4 and 3.3.5, to determine compliance with the applicable paragraph and drawing requirement.

4.2.3.3. Functional tests - The following tests shall be performed by the Government.

4.2.3.3.1 Ballistics - Twenty-one units shall be tested to determine compliance with the requirements of 3.4.1 through 3.4.6. The units shall be temperature conditioned as provided in Table II and 4.3.3.1. Failure of any unit to comply with the requirements of 3.4 shall be cause for rejection of the lot.

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TABLE II

Sample Size	Method	Temperature
7	Under load	-65° to -70°F
7	" "	65° to 75°F
7	" "	200° to 205°F

4.2.4 Inspection equipment - The examination and tests shall be made using equipment listed on IEL-1053600 9.

4.3 Test methods and procedures

4.3.1 Firing pin release - The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction, in conjunction with the fixture which registers the amount of pull.

4.3.2 Torque inspec tion -

4.3.2.1 A torque wrench set at the breakaway torque requirement specified in 3.3.1 shall be used in conjunction with the fixture to determine the adequacy of the joint.

4.3.2.2 A torque wrench set at the bottoming torque as specified in 3.3.2, 3.3.3, 3.3.4 and 3.3.5 shall be used in conjunction with an adapter to determine the adequacy of the joint.

4.3.3 Functional tests

4.3.3.1 Ballistics - The sample tested under the conditions as specified herein shall be mounted so that the weight shall be propelled horizontally and guided for the first two feet of travel on two rails which present a minimum of friction to the weight. Two strands of 35 pound tensile cord shall be used to give initial restraint to the propelled weight. The load cell, manifold, mounting fixture, ejector and projected weight shall be adjusted to a common center line which is parallel to the fixture. The aample shall be tested within ten minutes after removal from the conditioning chamber. When this time is exceeded the sample shall be reconditioned. A lanyard shall be attached between the second initiator and the projected weight in order to fire the initiator after two inches ± ½ inch of inner tube and projected weight travel.

A velocity and thrust recording system shall be used to determine these parameters of the moving weight at separation from the tube in the ballistic testing of any ejector under "load" condition.

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4.3.3.2 Temperature conditioning - An ejector assembly loaded with two dummy M91 Cartridges, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking- Level A - Packing and marking shall be in accordance with Drawings C9215469 and C9215470.

6. NOTES

6.1 Ordering data - Procurement documents should specify the title, number and date of this specification.

6.2 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 Load - The expected performance of the ejector, under conditions simulating those existing in parachute deployment shall be referred to as "load" conditions in the specification.

6.4 Explosive Hazard Classification of Material

6.4.1 Ejector, Parachute, Cartridge Actuated, XM233 Assembly

"Explosive Hazard Classification - Class 1"
"Storage Compatibility Grouping - Group B, E, or N"
"Department of Transportation -(DOT) Hazard Class - Class C"
"DOT Marking: Explosive Power Device - Class 'C'"
"Handle Carefully - Keep Fire Away"

Custodian:

ARMY MU

Preparing activity:

Army - MU (FA)

Project No. 1377-A479

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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